NetTrace Malware Traffic Analysis Report

PCAP: samples\suspicious\demo_beacon_http.pcap

Metric	Count
Packets	9
Dns Events	1
Http Events	1
Tls Events	0
Ftp Events	0
Flows	3
Iocs	3
Iocs Confirmed	3
Iocs Observed	0
Observed Artifacts	0
Findings	7
Critical	2
High	1
Medium	2
Low	1

Findings

CRITICAL - Local threat intel match

IOC matched the local known-bad intelligence list.
MITRE ATT&CK;: T1071.001 Application Layer Protocol: Web Protocols

```
Evidence: { "ioc_type": "domain", "ioc_value": "malware-test.example", "source":
"http_host", "packet_number": 9, "wireshark_filter": "frame.number == 9" }
```

CRITICAL - Local threat intel match

IOC matched the local known-bad intelligence list.
MITRE ATT&CK;: T1071.004 Application Layer Protocol: DNS

```
Evidence: { "ioc_type": "domain", "ioc_value": "xj3k9q2z7m1p0a8c.biz", "source":
"dns", "packet_number": 8, "wireshark_filter": "frame.number == 8" }
```

HIGH - Possible DGA domain

Domain structure has high entropy and weak language-like character patterns.
MITRE ATT&CK;: T1568.002 Dynamic Resolution: Domain Generation Algorithms

```
Evidence: { "domain": "xj3k9q2z7m1p0a8c.biz", "scored_label": "xj3k9q2z7m1p0a8c",
"entropy": 4.0, "dga_score": 0.981, "packet_number": 8, "wireshark_filter":
"frame.number == 8" }
```

MEDIUM - Possible executable/script download request

HTTP GET/HEAD requested an executable or script path.

MITRE ATT&CK;: T1071.001 Application Layer Protocol: Web Protocols

Evidence: { "url": "http://malware-test.example/payload.exe", "method": "GET", "packet_number": 9, "wireshark_filter": "frame.number == 9" }

MEDIUM - Possible beaconing behavior

Regular connection timing suggests command-and-control beaconing.

Evidence: { "src_ip": "10.0.0.5", "dst_ip": "203.0.113.66", "dst_port": 4444, "protocol": "TCP", "events": 7, "timing_events_truncated": false, "mean_interval_seconds": 10.0, "coefficient_of_variation": 0.0, "connection_count": 1, "observation_window_seconds": 60.0, "first_packet_number": 1, "packet_numbers_sample": [1, 2, 3, 4], "wireshark_filter": "frame.number in {1 2 3 4}" }

LOW - Connection to suspicious non-standard port

Traffic used a port commonly seen in malware labs, backdoors, or tunneling.

Evidence: { "dst_port": 4444, "protocol": "TCP", "dst_ip": "203.0.113.66", "src_ip": "10.0.0.5", "flow_count": 1, "distinct_destinations": 1, "distinct_sources": 1, "total_packets": 7, "total_bytes": 316, "top_peers": [{ "src_ip": "10.0.0.5", "dst_ip": "203.0.113.66", "packet_count": 7, "byte_count": 316 }], "first_packet_number": 1, "packet_numbers_sample": [1, 2, 3, 4, 5, 6, 7], "wireshark_filter": "frame.number in {1 2 3 4 5 6 7}" }

INFO - Command-line or automation HTTP client observed

HTTP request used a scripting/automation client (e.g. curl, wget, python-requests). This is common for legitimate developer, CI, and monitoring traffic -- treat as low-confidence context, not a standalone indicator, unless combined with other suspicious signals.

Evidence: { "host": "malware-test.example", "uri": "/payload.exe", "user_agent": "python-requests/2.28", "src_ip": "10.0.0.5", "dst_ip": "198.51.100.23", "packet_number": 9, "wireshark_filter": "frame.number == 9" }